

Full Paper

Effects of *Eppikahangeto*, a *Kampo* Formula, and *Ephedrae herba* Against Citric Acid-Induced Laryngeal Cough in Guinea PigsKiyoshi Minamizawa¹, Hirozo Goto^{1,*}, Yutaka Shimada^{1,2}, Katsutoshi Terasawa^{2,3}, and Akira Haji⁴¹Department of Japanese Oriental Medicine, Faculty of Medicine, and ²21st Century COE Program, University of Toyama, Toyama 930-0194, Japan³Department of Japanese Oriental Medicine, Graduate School of Medicine, Chiba University, Chiba 260-8670, Japan⁴Laboratory of Neuropharmacology, School of Pharmacy, Aichi Gakuin University, Nagoya 464-8650, Japan

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Abstract. To evaluate the efficacy of three common antitussive *Kampo* formulas, *eppikahangeto* (EPP), *bakumondoto* (BAK), and *shoseiryutogomakyokansekito* (SGM), a new cough model of guinea pig was used, which could specifically induce a laryngeal cough by microinjection of citric acid solution into the larynx. *Kampo* extract was dissolved in water and the animals were given access ad libitum for 3 days, and then the number of coughs during 10 min was counted. EPP extract decreased the number of coughs dose-dependently (0.3% extract, $-22.9 \pm 6.6\%$, $P < 0.01$; 1.0% extract, $-32.4 \pm 5.5\%$, $P < 0.01$). BAK extract and SGM extract had no significant effect. Intraperitoneal injection of codeine (60 mg/kg) also decreased the number of coughs ($-36.1 \pm 9.1\%$, $P < 0.05$). Furthermore, *Ephedrae herba* (EH) extract reduced the number of coughs ($-18.3 \pm 6.0\%$, $P < 0.05$), but the extract of EPP without EH did not. These results suggest that EPP has an antitussive effect against laryngeally-induced cough in guinea pigs, and the crucial herbal medicine is EH.

Keywords: *Kampo* medicine, *eppikahangeto*, *Ephedrae herba*, antitussive effect, laryngeal cough

Introduction

Coughing is the most common reason for visiting clinics (1). Coughing is basically a beneficial and important defensive reflex against irritants or foreign materials that attempts to keep the respiratory tract clear. However, too frequent coughing diminishes the patient's comfort, and chronic persistent coughing leads to a deterioration in the quality of life because of sleep disturbance, exhaustion of physical energy, rib fracture, and so on (2, 3). Thus, antitussive agents are frequently prescribed, and traditional Chinese/Japanese (*Kampo*) formulas are also used for antitussive purpose in Japan. While the popular antitussive agents such as codeine and dextromethorphan are used regardless of the nature of the cough, each *Kampo* formula has strict indications for the character of the cough and the patient's condition according to the traditional concept of *Kampo* medicine.

For example, *eppikahangeto* (EPP) is administered to those who are suffering from a severe spasmodic cough accompanied by emesis, red face, and feeling of heat in the entire body (4). *Bakumondoto* (BAK) is applied for those who are suffering from dry cough, hoarseness, and difficulty in coughing out sputum (5); *Shoseiryuto* is indicated for the patients suffering from a cold accompanied by coughing, sneezing, serous foamy phlegm, and serous rhinitis (6); and *makyokansekito* is used for those with a severe cough, thirst, sticky glutinous phlegm, and feeling of heat in the entire body (7). In Japan, a combination formula of *shoseiryuto* and *makyokansekito*, *shoseiryutogomakyokansekito* (SGM), has often been used empirically for treatment of a severe cough. By using an experimental animal model, Miyata (8) reported the suppressive effects of EPP, BAK, and SGM against a cough induced by inhalation of citric acid. However, thereafter, the action mechanism of the antitussive effects of these formulas has not been well investigated.

To investigate the antitussive effect of drugs, one

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of the most usual techniques to elicit a cough in unanesthetized unrestrained animals is inhalation of tussigenic agents. However, there are three principal shortcomings to this procedure. First, as this maneuver stimulates the entire respiratory tract from nose to peripheral bronchus, not only coughing but also other respiratory reflexes such as sneezing, which are hardly distinguishable from coughing by sound or ventilation waveform, might be elicited. Second, the number of coughs may vary due to the respiratory condition such as diving reflex or deep inhalation following frequent coughing. Third, this maneuver cannot be repeated on the same animal because of tachyphylaxis (9–11). Recently, a new coughing model of guinea pigs able to overcome these disadvantages was developed by Tanaka and Maruyama (12). In this model, a small amount of citric acid solution is microinjected into the larynx. Since this stimulation is localized at the larynx, specifically causing laryngeal coughing and not influenced by the respiratory condition, the cough response can be easily differentiated from other responses, and it can be evaluated quantitatively. Furthermore, in this model, tachyphylaxis is not induced. The most common sites known to produce cough reflex are the larynx and the bronchial bifurcation (13–15), and it has been suggested that there are some differences in the regulation and response to antitussives between laryngeal cough and bronchial cough (16–19). Therefore, in the present study, we evaluated the antitussive effects of three *Kampo* formulas, EPP, BAK, and SGM, which are often administered to patients suffering from a cough, against citric acid-induced laryngeal cough by using this new animal model.

Materials and Methods

Preparation of Kampo extract

Extracts of *Kampo* formulas, EPP, BAK, SGM, were prepared in our laboratory. These formulas consisted of crude drugs in ratios based on the traditional regimens as shown in Table 1. Extracts of *Ephedrae herba* (EH) alone and EPP without EH (EPPw/oEH) were also prepared. All crude drugs, identified to genus and species levels, were purchased from Tochimoto Pharmaceuticals (Osaka). The mixtures of crude drugs or EH alone (total 100 g) were extracted with boiling water (500 ml) for 50 min. The solution was filtered and centrifuged at $10,000 \times g$ for 30 min, and the supernatant was then converted to freeze-dried powder. The yields of EPP extract (EPPE), BAK extract (BAKE), SGM extract (SGME), EH extract (EHE), and EPPw/oEH extract (EPPw/oEHE) were 9.9%, 16.2%, 7.8%, 8.2%, and 11.2%, respectively.

Three-dimensional high-performance liquid chromatography (3D-HPLC) analysis of EPPE

3D-HPLC of EPPE is shown in Fig. 1. The HPLC conditions were as follows: EPPE (0.5 g) was dissolved in 50 ml of 50% methanol under ultrasonication for 30 min. The solution was filtered with a membrane filter (0.45 μ m, DISMIC 13HP; Advantec, Tokyo) and then subjected to HPLC analysis. HPLC equipment was controlled with LC-10A (Shimizu, Tokyo) using a TSK-GEL ODS-80TS column (4.6 $\phi \times 250$ mm), eluting with solvents (I) 0.05 M AcOH-AcONH₄ (pH 3.6) and (II) CH₃CN. A linear gradient of 90% I and 10% II changing over 60 min to 0% I and 100% II was used. The flow rate was controlled with an SLC-10AD pump at 1.0 ml/min. The eluate from the column was monitored, and three-dimensional data were processed by SPD-M10A diode array detectors (Shimadzu, Kyoto). All assigned peaks were identified by co-injection test with authentic samples and compared with UV spectral data.

Animals

Male Hartley guinea pigs purchased from Japan SLC, Inc. (Hamamatsu) weighing 400–600 g were used for the experiments. The animals were placed individually into plastic cages and housed in an animal room at a temperature of $24 \pm 2^\circ\text{C}$ and a 12-h light-dark cycle. Animals were allowed free access to standard laboratory food. All experimentation was performed in accordance with the Guide Principles for the Care and Use of Laboratory Animals approved by the Japanese Pharmacological Society and also approved by the Committee on Animal Experimentation, University of Toyama.

Surgical procedures

The experiment was performed according to the procedure of Tanaka and Maruyama (12). A guinea pig was anesthetized with an intramuscular injection of a mixed solution of ketamine hydrochloride (80 mg/kg; Wako Pure Chem. Ind., Osaka) and xylazine (8 mg/kg; Sigma, St. Louis, MO, USA). The animal was fixed on a bed in supine position. The trachea was exposed surgically and a polyethylene tube (inside diameter, 0.5 mm; outside diameter, 0.9 mm; length, 17.5 cm; SP-35; Natume Seisakusho Co., Ltd., Tokyo) was inserted between the seventh and eighth cartilage. The tip of the tube was advanced 15-mm toward the larynx to place it just under the larynx. Then the tube was fixed with ligation to the accompanying sternohyoid muscle, and the other end was led out from the dorsum of the neck through a subcutaneous tunnel. All experiments were started after one-week recovery. At the end of all experiments, a lethal dose of pentobarbital sodium

Table 1. Crude drugs composing *eppikahangeto*, *bakumondoto*, and *shoseiryutogomakyokansekitō*

Latin name	Japanese name	Botanical name (Family name)	Origin	Ratio (g)
<i>Eppikahangeto</i>				
<i>Ephedrae herba</i>	<i>Mao</i>	<i>Ephedra sinica</i> STAPF (Ephedraceae)	China	6
<i>Gypsum fibrosum</i>	<i>Sekko</i>	CaSO ₄ • 2H ₂ O [#]	China	8
<i>Pinelliae tuber</i>	<i>Hange</i>	<i>Pinellia ternata</i> BREITENBACH (Araceae)	China	6
<i>Zizyphi fructus</i>	<i>Taiso</i>	<i>Zizyphus jujuba</i> MILLER (Rhamnaceae)	China	3
<i>Glycyrrhizae radix</i>	<i>Kanzo</i>	<i>Glycyrrhiza uralensis</i> FISCHER (Leguminosae)	China	2
<i>Zingiberis rhizoma</i>	<i>Shokyo</i>	<i>Zingiber officinale</i> ROSCOE (Zingiberaceae)	China	1
<i>Bakumondoto</i>				
<i>Ophiopogonis tuber</i>	<i>Bakumondo</i>	<i>Ophiopogon japonicus</i> KER-GAWLER (Liliaceae)	China	10
<i>Oryzae semen</i>	<i>Kobei</i>	<i>Oryza sativa</i> LINNE (Gramineae)	Japan	5
<i>Pinelliae tuber</i>	<i>Hange</i>	<i>Pinellia ternata</i> BREITENBACH (Araceae)	China	5
<i>Zizyphi fructus</i>	<i>Taiso</i>	<i>Zizyphus jujuba</i> MILLER (Rhamnaceae)	China	3
<i>Glycyrrhizae radix</i>	<i>Kanzo</i>	<i>Glycyrrhiza uralensis</i> FISCHER (Leguminosae)	China	2
<i>Ginseng radix</i>	<i>Ninjin</i>	<i>Panax ginseng</i> C. A. MEYER (Araliaceae)	South Korea	2
<i>Shoseiryutogomakyokansekitō</i>				
<i>Ephedrae herba</i>	<i>Mao</i>	<i>Ephedra sinica</i> STAPF (Ephedraceae)	China	3
<i>Peoniae radix</i>	<i>Shakuyaku</i>	<i>Peonia lactiflora</i> PALLAS (Paeoniaceae)	Japan	3
<i>Zingiberis siccatur rhizoma</i>	<i>Kankyo</i>	<i>Zingiber officinale</i> ROSCOE (Zingiberaceae)	China	3
<i>Glycyrrhizae radix</i>	<i>Kanzo</i>	<i>Glycyrrhiza uralensis</i> FISCHER (Leguminosae)	China	3
<i>Cinnamomi cortex</i>	<i>Keihi</i>	<i>Cinnamomum cassia</i> BLUME (Lauraceae)	China	3
<i>Asiasari radix</i>	<i>Saishin</i>	<i>Asiasarum sieboldi</i> F. MAEKAWA (Aristolochiaceae)	North Korea	3
<i>Schizandrae fructus</i>	<i>Gomishi</i>	<i>Schizandra chinensis</i> BAILLON (Magnoliaceae)	China	3
<i>Pinelliae tuber</i>	<i>Hange</i>	<i>Pinellia ternata</i> BREITENBACH (Araceae)	China	6
<i>Armeniaca semen</i>	<i>Kyonin</i>	<i>Prunus armeniaca</i> LINNE (Rosaceae)	China	4
<i>Gypsum fibrosum</i>	<i>Sekko</i>	CaSO ₄ • 2H ₂ O [#]	China	10

[#]Mineral source.

(100 mg/kg; Dainippon Pharmaceutical Co., Ltd., Osaka) was injected intraperitoneally for sacrifice.

Experimental schedule

Before drug administration, pre-treatment (control) trials were carried out. Thereafter, drinking water including *Kampo* extract at a weight/solvent volume concentration (w/v) of 0.1%–1.0% was given access ad libitum to animals for 3 days, and then post-treatment trials were performed. After a 7-day drug washout period, pre- and post-treatment trials were repeated two more times. The same animal was set up to receive a different treatment in each trial. As a positive control, some of the animals were randomly assigned to be given 60 mg/kg of codeine (Takeda Chemical Ind., Ltd., Osaka) intraperitoneally 15 min before the laryngeal stimulation instead of *Kampo* extract treatment.

Plethysmographic measurement

The cough was evaluated by whole-body plethysmo-

graphy under unrestrained, unanesthetized condition. The animal was placed in a sealed chamber (5 L) into which continuous airflow (2 L/min) was sent. The implanted polyethylene tube of the animal was connected to an extension tube and led outside through a pinhole in the chamber. After 15 min, 20 µl of 7.5% citric acid (Wako) was microinjected for 5 s directly into the larynx through the tube to elicit coughing. A microinjection of 20 µl of physiological saline (vehicle) alone did not induce cough by this procedure. Pressure changes derived from breath-to-breath ventilation were measured with a differential pressure transducer (TP-602T; Nihon Kohden, Tokyo), and sounds were recorded with a microphone placed inside the chamber for 10 min. The data were processed (MacLab/4s; AD Instruments, Castle Hill, Australia) and then monitored on a computer display and stored on a hard disk via signal processing software (PowerLab Chart v5, AD Instruments).

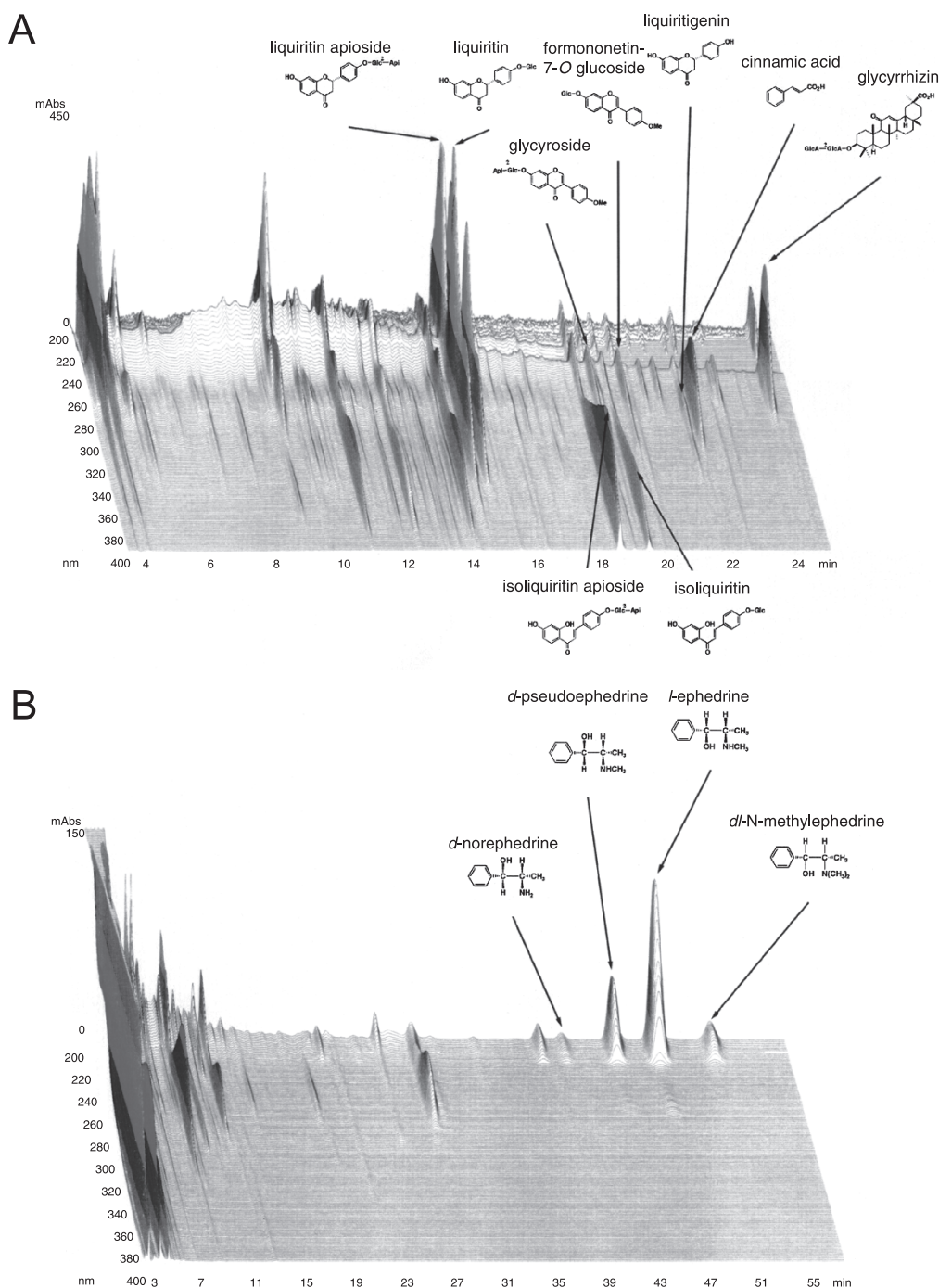


Fig. 1. Three-dimension high-performance liquid chromatography (3D-HPLC) profile of the water solution of *eppikahango* extract (EPPE). A: 3D-HPLC fingerprint of EPPE. B: 3D-HPLC fingerprint of ephedra alkaloids in EPPE.

Determination of cough reflex

Cough reflex was determined by ventilation wave-form pattern and sound according to the report of Tanaka and Maruyama (12). The criteria for coughing employed in this study were I) a deep initial inspiration followed by rapid large expiration, II) the expiration amplitude more than 200% of normal expiration, and III) simultaneous sound.

Data analysis and statistics

The numbers of coughs in the control trial and post-treatment trial were counted for 10 min from just after laryngeal stimulation on the computer screen, and the percent decrease in the number of coughs in the post-treatment trial from that in the control trial was calculated. Data were presented as the mean $\pm$ S.E.M. Statistical analysis of the differences in the numbers of

coughs in the control trials and post-treatment trials was performed by Student's paired *t*-test. *P*<0.05 was considered to be statistically significant.

Results

Cough reflex pattern

Whole-body plethysmography showed regular small

inspiratory-expiratory waveforms reflecting regular breathing. Microinjection of citric acid into the larynx elicited coughing (Fig. 2A), characterized by a waveform pattern having deep inspiration followed by rapid large expiration simultaneously with sound. This biphasic pattern was the same as that previously reported by Piiria and Sovijarvi (20) (Fig. 2B). Not single but double or triple cough reflexes sometimes occurred

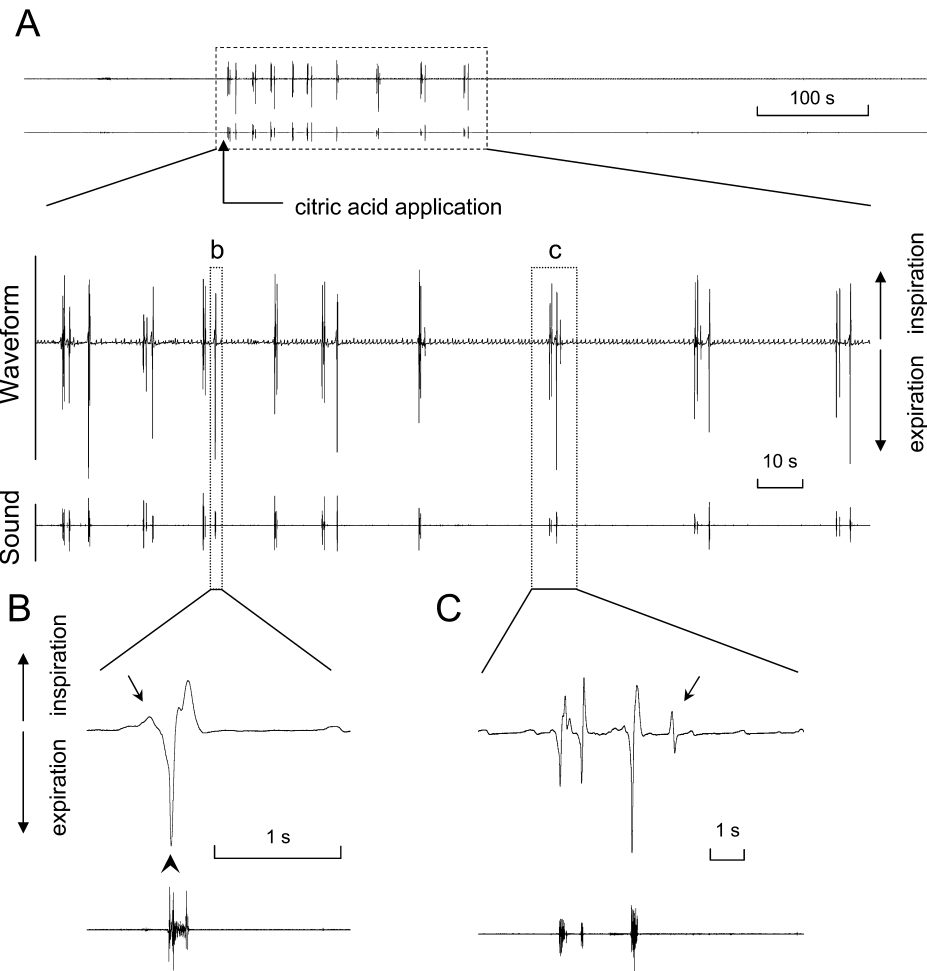


Fig. 2. Representative plethysmographic traces of ventilation waveform (Waveform) and sound volume (Sound). A: Cough reflexes are evoked after citric acid application, especially in the first few minutes. Large flow deflections in waveform represent cough reflexes. B: Magnified view of a typical cough response (b in Fig. 2A), determined by the waveform pattern with deep inspiration (oblique arrow) followed by rapid large expiration (arrowhead) with simultaneous sound. C: Magnified view of a series of continual coughs (c in Fig. 2A). Oblique arrow exhibits an example of a sigh, showing a deflection of airflow but without sound.

Table 2. Water volume and amount of extract taken by guinea pigs

	Water intake (mL/day)	Extract intake (mg/day)
0.1% <i>eppikahangeto</i> extract	119.4 ± 7.3	119.4 ± 7.3
0.3% <i>eppikahangeto</i> extract	114.0 ± 8.6	341.9 ± 25.8
1.0% <i>eppikahangeto</i> extract	77.3 ± 5.1	773.3 ± 51.2
1.0% <i>bakumondoto</i> extract	138.2 ± 9.0	1382.0 ± 89.8
1.0% <i>shoseiryutogomakyokansekitto</i> extract	83.5 ± 5.7	834.8 ± 56.5
0.3% <i>Ehedralae herba</i> extract	77.4 ± 9.1	232.3 ± 27.3
1.0% <i>eppikahangeto</i> without <i>Ehedralae herba</i> extract	111.8 ± 2.3	1117.5 ± 23.3

Data represent the mean ± S.E.M. (n = 24 – 48).

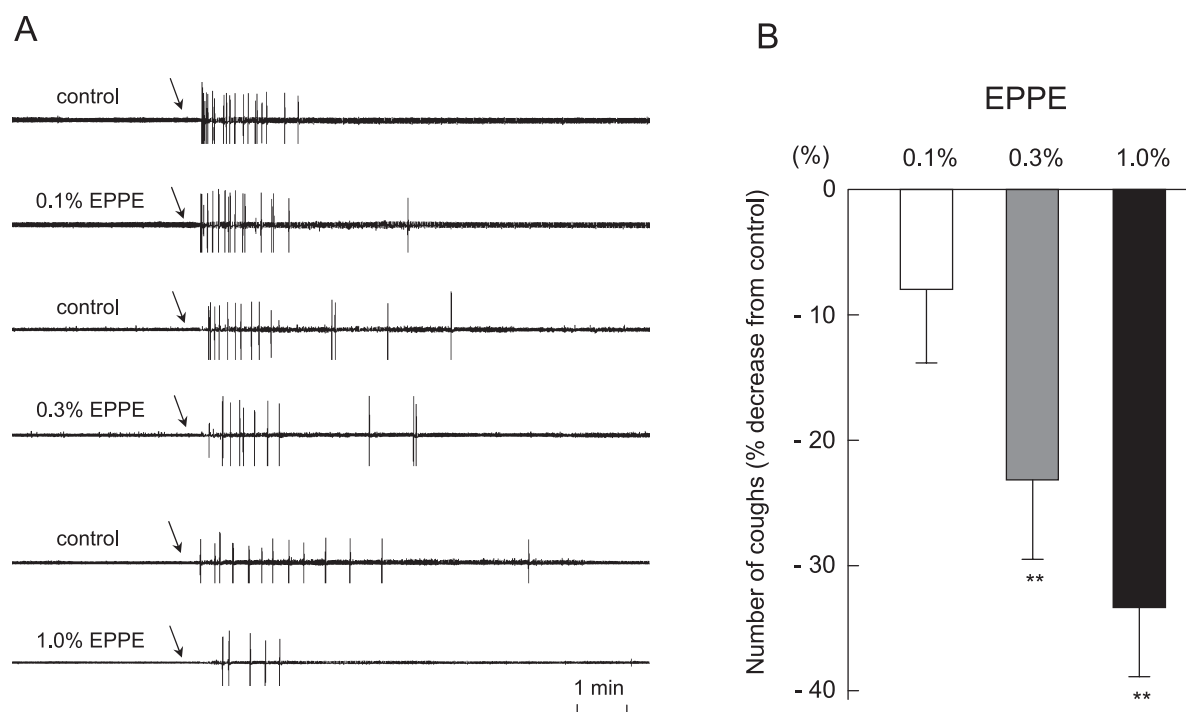


Fig. 3. Effect of *eppikahangeto* extract (EPPE) against citric acid-induced laryngeal cough in guinea pigs. A: Representative plethysmographic traces of ventilation waveform before (control) and after 3-day 0.1%, 0.3%, and 1.0% EPPE-treatments. Oblique arrows indicate the time point of citric acid (7.5%, 20 μ l) microinjection into the larynx. B: Number of coughs after 3-day 0.1%, 0.3%, and 1.0% EPPE-treatments (percent decrease from control). Each bar represents the mean $\pm$ S.E.M. (n = 15 – 16). ** P < 0.01 (compared to control).

during the measurement. Waveforms with a pattern of a relatively small but larger than normal expiration were discarded because they were soundless (Fig. 2C). During the 10-min observation, cough reflexes occurred frequently in the first few minutes after laryngeal stimulation. The average number of coughs in 10 min in the control trials was 22.6 ± 0.9 (n = 103).

Effects of *Kampo* extracts

Water volume and the amount of extract taken by the animals are shown in Table 2. As taste and smell differ among the extracts, the amount of extract taken varied even at the same concentration. However, a dose-dependency for the amount of taken extract was seen among 0.1%, 0.3%, and 1.0% EPPE solutions.

EPPE decreased the number of coughs dose-dependently. Typical examples of ventilation pressure waveforms in the animals treated with EPPE are shown in Fig. 3A and the percent decrease in the number of coughs is shown in Fig. 3B. On the other hand, 1.0% BAKE did not decrease the number of coughs, and 1.0% SGME showed a decreasing tendency without significance ($P = 0.075$). Intraperitoneal injection of codeine (60 mg/kg) reduced the number of coughs significantly (Fig. 4).

To clarify which crude drug of those comprising EPP is crucial for the antitussive effect, we focused on EH because it is one of the herbal medicines contained in both EPP and SGM, and it has a long history of use for the treatment of respiratory disorders (21, 22). Treatment with 0.3% EHE significantly decreased the number of coughs, whereas EPPw/oEHE did not (Fig. 5).

Discussion

We evaluated the antitussive effect of *Kampo* formulas using a new cough model of guinea pigs. This model could induce a laryngeal cough selectively, and the evoked cough response was qualitatively distinguishable from other responses and could be quantitatively evaluated (12). The present study demonstrated that EPPE possessed a significant antitussive effect in a dose-dependent fashion. SGME showed a tendency to suppress the cough response, whereas BAKE had no antitussive effect.

Previously, Miyata (8) reported that all of three *Kampo* formulas, EPP, BAK, and SGM, had antitussive effects against coughing induced by inhalation of citric acid aerosol, with the effect of SGM being the strongest.

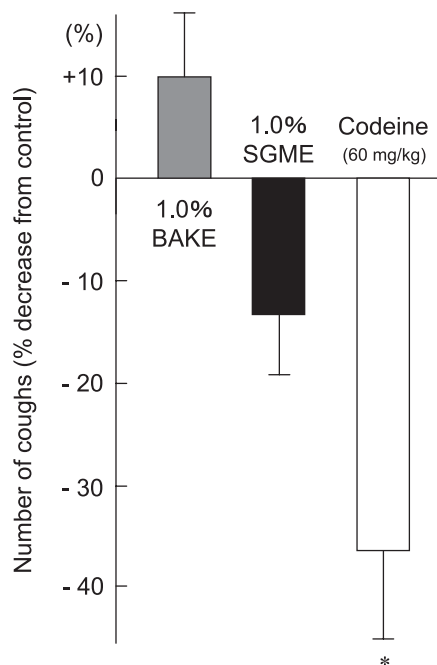


Fig. 4. Effects of 1.0% *bakumondoto* extract (BAKE), 1.0% *shoseiryutogomakyokansekito* extract (SGME), and codeine against citric acid-induced laryngeal cough in guinea pigs. Number of coughs after 3-day 1.0% BAKE- and 1.0% SGME-treatments, and 15 min after intraperitoneal codeine (60 mg/kg) injection (percent decrease from pre-treatment trial (control)). Each bar represents the mean $\pm$ S.E.M. (1.0% BAKE, $n = 16$; 1.0% SGME, $n = 16$; codeine, $n = 6$). * $P < 0.05$ (compared to the control).

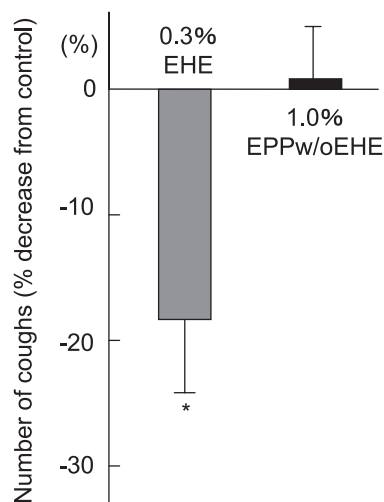


Fig. 5. Effects of 0.3% *Ephedrae herba* extract (EHE) and 1.0% *eppikahangeto* without *Ephedrae herba* extract (EPPw/oEHE) against citric acid-induced laryngeal cough in guinea pigs. Number of coughs after 3-day 0.3% EHE- and 1.0% EPPw/oEHE-treatments (percent decrease from pre-treatment trial (control)). Each bar represents the mean $\pm$ S.E.M. ($n = 8$). * $P < 0.05$ (compared to the control).

This maneuver stimulates not only the upper but also the lower respiratory tract, resulting in evoking both laryngeal cough and tracheobronchial cough. Coughing is induced by various tussigenic stimuli to the airway. The sites known to be most sensitive for evoking the cough reflex are the larynx and tracheal bifurcation, both innervated by vagal branches (13 – 15). Afferent signals from airway sensory receptors, for example, C-fiber receptors, rapidly adapting receptors, and slowly adapting stretch receptors, activated initially or secondarily by various tussigenic stimuli, terminate in the nucleus tractus solitarius through the vagus nerve (23, 24). The interactions among these components of the cough pathway have not been fully elucidated (25). However, there are some differences in the responses to stimuli between the laryngeal and tracheobronchial coughs (16 – 18). Furthermore, differences in response to anti-tussive agents between them have also been reported (19). The method used in the present study is capable of evoking only the laryngeal cough. The difference between the results of Miyata (8) and our present study might reflect the antitussive character of the *Kampo* medicine and/or the difference of the methods. Tanaka and Maruyama described that the microinjection of citric acid stimulates both A δ -fiber and C-fiber (12), whereas its inhalation stimulates only C-fiber (26). Therefore, there is the possibility that EPP may influence A δ -fiber rather than C-fiber. It is reported that BAK has a strong antitussive effect in the bronchitis model of guinea pig, an accelerating effect on tracheal mucociliary clearance, and an inhibitory effect against airway hypersensitivity induced by ozone exposure (27 – 29). Therefore, we consider that BAK might act on the bronchus rather than the larynx.

EH is the one of the herbal medicines comprising EPP and SGM, but is not included in BAK (Table 1). In the present study, both EPPE and EHE exhibited an antitussive effect, but EPPw/oEHE did not. Therefore, the antitussive effect of EPP against the laryngeal cough is considered to be due mainly to EH. Traditionally, EH has been used for the treatment of respiratory disorders. Ephedra alkaloids, the most famous ingredients of EH, for example, ephedrine, pseudoephedrine, and methylephedrine, are well known to be a sympathomimetic agonist at both α - and β -adrenergic receptors and are used for this purpose in bronchodilators (21, 22). It has also been reported that ephedrine and methylephedrine have an antitussive effect in humans (30) and animal models (31, 32). Nevertheless, the antitussive effect of EH itself and its alkaloids has not been thoroughly investigated. According to the weight ratio of EH in EPP (approx. 23%), it is considered that 0.3% EHE solution contained more ingredients derived

from EH than not only 0.3% EPPE and but also 1.0% EPPE solution. Nevertheless, the antitussive effect of 0.3% EHE was likely to be weaker than that of 1.0% EPPE and similar to that of 0.3% EPPE. This seems to indicate that EH is crucial, but that other crude drugs also play an important role in the antitussive effect of EPP.

In conclusion, the present study revealed that EPP had antitussive effects against the citric acid-induced laryngeal cough in guinea pigs, and the crucial herbal medicine was EH. Further studies are needed to clarify the active component and reveal greater details of the action mechanism of the antitussive effects of EPP and EH.

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